

Chapter 6: Inferential Methods for a Single Numerical Variable

As seen in earlier chapters, research hypotheses involving either a single categorical variable or two categorical variables require us to test claims about population proportions. When a research question involves a single numerical variable, however, we end up testing claims about the population mean (or average) based on a sample. In this chapter, we will consider inferential methods (hypothesis tests and confidence intervals) for the population mean of a single numerical variable. Consider the following example.

Notation for Means

Sample (these are statistics)

- $\bar{x}$ is the sample mean (think of this as $\hat{p}$ from proportions)
- s is the sample standard deviation

Population (these are parameters)

- μ is the population mean (think of this as π from proportions)
- σ is the population standard deviation

Example 6.1: Time Perception Impaired by Nicotine Withdrawal

A study conducted by researchers at Pennsylvania State University investigated whether time perception, a simple indication of a person's ability to concentrate, is impaired during nicotine withdrawal. The study results were presented in the paper *Smoking Abstinence Impairs Time Estimation Accuracy in Cigarette Smokers* (Klein, Corwin, and Stine 2003). After a 24-hour smoking abstinence, 20 daily smokers were asked to estimate how much time had passed during a 45-second period. Suppose the resulting data on perceived elapsed time (in seconds) were summarized and visualized as shown below (these results are artificial but are similar to the actual findings).

Research Question: Is there evidence the mean perceived elapsed time for all smokers suffering from nicotine withdrawal is significantly greater than the actual 45 seconds?

```
# A tibble: 6 × 4
  status sex age time_passed
  <chr>   <chr> <dbl>   <dbl>
1 withdrawal female 33.0     49.0
2 withdrawal male 42.6     47.9
3 withdrawal female 41.8     52.5
4 withdrawal female 37.5     54.0
5 withdrawal male 38.0     41.7
6 withdrawal male 48.0     50.7
```

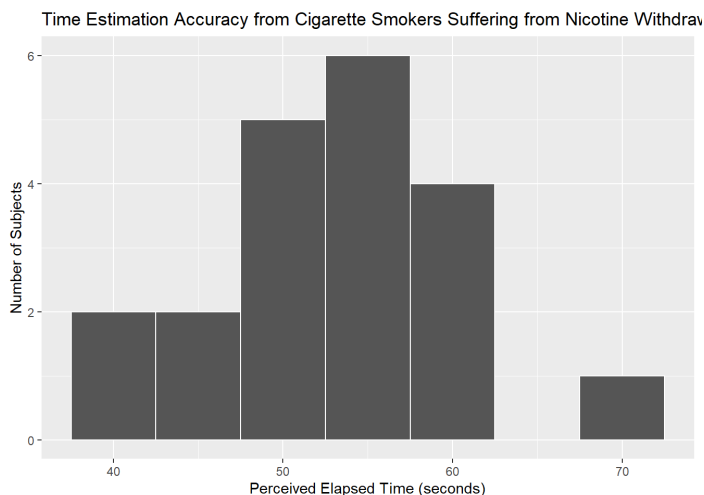
We can summarize the data with:

```
favstats( ~ time_passed, data = nicotine)
```

```
      min      Q1  median      Q3      max      mean      sd n missing
39.11611 48.74039 53.25595 56.65965 69.36787 52.65206 7.268201 20      0
```

and visualize the data with:

```
ggplot(data = nicotine,
       mapping = aes(x = time_passed)) +
  geom_histogram(binwidth = 5) +
  labs(title = "Distribution of our data",
       y = "Number of Subjects",
       x = "Perceived Elapsed Time (seconds)")
```



1. Identify the following in context of the scenario:

- Population: ALL smokers suffering from nicotine withdrawal
- Sample: $n = 20$ smokers suffering from nicotine withdrawal in our study
- Variable of interest: perceived time passed (time_passed)
- Data Type: numeric

2. What is the mean of the observed data? The standard deviation?

- $\bar{x} = 52.65$ (sample mean)
- $s = 7.27$ (sample standard deviation)

3. If another sample of $n = 20$ subjects were obtained, would these new subjects have a mean exactly the same as the mean from this sample? Why or why not?

No, you would have a different set of 20 individuals.

4. Given your answer to the previous question, do you think it is appropriate to use only this sample mean to make inferences about the mean perceived elapsed time being greater than 45 seconds in the population of all smokers subjected to nicotine withdrawal? Explain.

No, we need to understand the sample to sample variability of the sample mean.

The Distribution of the Sample Mean

The sample mean is a random quantity that changes from sample to sample. This randomness gives the sample mean its own distribution, called the distribution of the sample mean, which tells us:

- The possible values the sample mean can assume.
- How often each value will occur.

Understanding this distribution is key to making decisions about the population mean for a single numerical variable. Let's explore how sample means behave by simulating repeated samples.

Exploring the Distribution of the Sampling Mean

We will use an applet (<https://emily-robinson.shinyapps.io/06-CLT-app/>) to simulate drawing random samples from a population and observe the distribution of sample means.

The goal is to simulate multiple samples of $n = 20$ drawn from a hypothetical population of smokers experiencing nicotine withdrawal.

Step 1: Population Parameters We begin by setting up a hypothetical population of **all** smokers suffering from nicotine withdraw.

In this simulation study, what are the following parameters for your population?

$\mu = 45$ – Where does this value of 45 seconds come from? *yours might differ slightly*

If there is no impairment, we expect there to be a mean of $\mu = 45$ seconds perceived to pass.

$\sigma = 7.3$

What does each dot on this plot represent? **One smoker with nicotine withdrawal in the entire population.**

Step 2: Repeated Samples In practice, we usually don't know the true population mean (μ). Researchers often take random samples to estimate this mean. In our case, we'll simulate multiple random samples of 20 subjects.

- *Sample 1:* Click "Draw Samples". Record the sample mean and sample standard deviation:

$\bar{x}_{sample1} = 45.9$ (remember yours will differ due to randomness)

$s_{sample1} = 6$

What does each dot on this plot represent? **One smoker suffering nicotine withdrawal in the first sample of 20 individuals.**

- *Sample 2:* Click "Draw Samples" again. Record the new values:

$\bar{x}_{sample2} = 45.4$

$s_{sample2} = 7.4$

- *Sample 3:* Click "Draw Samples" once more. Record:

$$\bar{x}_{sample3} = 44.4$$

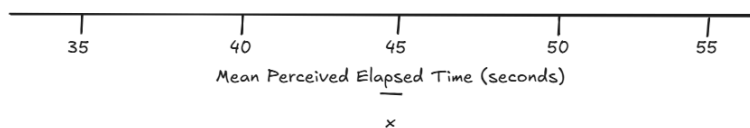
$$s_{sample3} = 8.4$$

Compare: How do the means and standard deviations from these three samples differ? **They are all different because we have a different set of 20 individuals in each sample. Notice the means are all kinda close to 45.**

Step 3: Distribution of Sample Means: Each of these sample means is a single point from the *distribution of sample means*. Let's now visualize the distribution of sample means:

Add a dot on the plot for each sample mean from the three samples you took above.

Distribution of Sample Means under assumption
the population mean $\mu = 45$ seconds
(i.e., the null is true)



Note this will start to be centered at 45 seconds (aka assumed population mean under no impairment) – null true!

Next: To get a better sense of this distribution, increase the “Number of Samples” to 100 and hit “Draw Samples” to generate more sample means. Record the overall mean and standard deviation from the “Distribution of Sample Means”.

- Distribution Mean = 45 **Same as population mean μ !**
- Distribution SD = 1.7 **See CLT below! This is roughly $7.3/\sqrt{20} = 1.63$**

What does each dot on this plot represent? **The mean perceived time passed for one sample of 20 smokers suffering nicotine withdrawal under the assumption there is no impairment ($\mu = 45$ seconds).**

Step 4: Hypothesis Testing The researchers hypothesized that the mean perceived elapsed time for smokers during nicotine withdrawal was greater than 45 seconds. Let's formally define the hypotheses:

H_0 : The mean perceived elapsed time for *all* smokers suffering from nicotine withdrawal is equal to 45 seconds. **(No impairment)**

H_A : The mean perceived elapsed time for *all* smokers suffering from nicotine withdrawal is *greater* than 45 seconds. **(impairment)**

The “Distribution of Sample Means” assumes the null hypothesis is true. It shows what we expect sample means to look like if the true population mean really is 45 seconds.

Step 5: Observed Data Recall, in the actual research study, the mean perceived elapsed time for 20 subjects was 52.65 seconds. Sketch this observed value on the “Distribution of Sample Means” you generated.

Estimate the p-value: Based on the simulation, what is the probability of obtaining a sample mean of $\bar{x} = \$52.65$ seconds or greater, assuming the null hypothesis is true?

Estimated p-value = <0.0001

Step 6: Understanding the Central Limit Theorem (CLT)

Rather than always using simulations, statisticians rely on a theoretical result called the *Central Limit Theorem (CLT)*. The CLT allows us to make inferences about the population mean without needing to take repeated samples.

The Central Limit Theorem (CLT)

The Central Limit Theorem (CLT) allows us to approximate the *distribution of sample means* as normal, provided the sample size is large enough.

- **Mean of Sample Means:** The *distribution of sample means* will be centered around the population mean μ . This means that, on average, the sample mean will be a good estimate of the population mean.
- **Standard Error (SE):** The standard deviation of the *distribution of sample means* is called the standard error. It tells us how much the sample means tend to vary from the population mean:

$$SE = \frac{\sigma}{\sqrt{n}}$$

- **When is the CLT valid?** For most situations, the CLT applies well if the sample size is 30 or larger. If the sample size is smaller than 30, the population should be normally distributed. For populations that are strongly skewed or have outliers, larger samples may be necessary for the CLT to hold.

- Check the Shape: Does the "Distribution of Sample Means" look approximately normal? **Yes! Symmetric and bell shaped (one peak)**
- Recall from Step 1: The population parameters were $\mu = 45$ and $\sigma = 7.3$
- Calculate the Standard Error (SE): We call the standard deviation of the distribution of sample means the standard error. It represents how much variability we expect in the sample means if we repeatedly take samples from the population.

$$SE = \frac{\sigma}{\sqrt{n}} = 7.3/\sqrt{20} = 1.63 \approx 1.7 \text{ from sampling distributions}$$

Step 7: Comparing Simulation Results and Theory

Compare the values from your simulation with the theoretical values based on the CLT:

- From Step 3: Mean = 45, SD = 1.7
- From Step 6: SE = 1.63

How does simulated standard deviation compare to the theoretical SE?

About the same! Note SE (standard error) is the fancy name for the standard deviation of the sampling distribution of means.

Key Statistical Concepts

The Central Limit Theorem allows us to estimate the population mean using a single sample, knowing that:

1. The distribution of sample means will be approximately normal if the sample size is large enough.
2. The sample means will be centered around the true population mean, μ .

3. The variability of the sample means is determined by the standard error $\frac{\sigma}{\sqrt{n}}$, which decreases as the sample size increases.

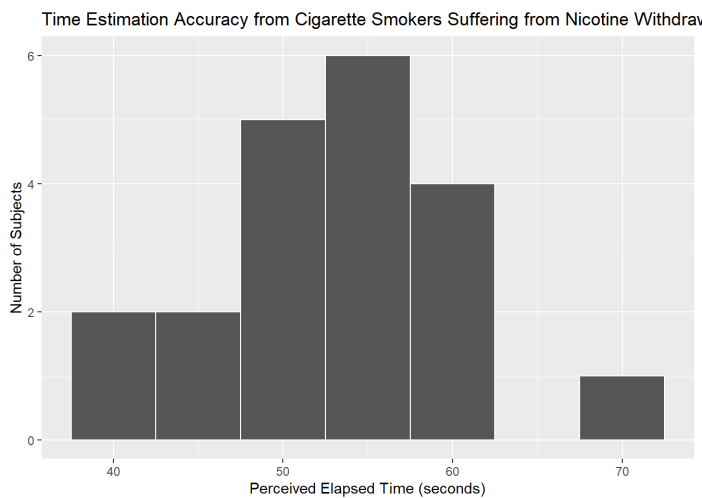
You can access a fun applet for exploring Sampling Distributions and the CLT further at https://onlinestatbook.com/stat_sim/sampling_dist/index.html.

In the next section, we will use the CLT to perform hypothesis tests using a procedure called the t-test.

The t-test for a Single Population Mean

Back to **Example 6.1**: Recall that the researchers wanted to show that the mean perceived elapsed time for smokers suffering from nicotine withdrawal was greater than the actual 45 seconds that had elapsed. The data collected in the study were summarized as follows:

min	Q1	median	Q3	max	mean	sd	n	missing
39.11611	48.74039	53.25595	56.65965	69.36787	52.65206	7.268201	20	0



Setup

1. What is the parameter of interest?

μ : The population mean perceived time passed for all daily smokers suffering nicotine withdrawal.

The population mean [variable of interest] for [population].

2. Set up the null and alternative hypotheses

H_0 : The population mean perceived time passed for all daily smokers suffering nicotine withdrawal is equal to 45 seconds. (no impairment)

H_A : The population mean perceived time passed for all daily smokers suffering nicotine withdrawal is greater than 45 seconds. (impairment)

In Symbols:

- $H_0 : \mu = 45$
- $H_A : \mu > 45$

Find the t-statistic and the p-value

To determine whether or not the distance between μ (the hypothesized population mean, null value) and (the mean from our observed sample) is larger than what we would expect by random chance, we will use the following statistic:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{52.65 - 45}{7.27/\sqrt{20}} = 4.71$$

recall z-scores! The mean of our observed data is almost 5 sd above the expected mean (based on the sampling distribution of the mean under the assumption the null is true $\mu = 45$), therefore our mean of 52.65 seconds is probably an outlier → reject the null.

Why use this statistic? Because this quantity measures the position of our observed sample mean on the null model, just like the Z-score discussed in the previous chapter.

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{\text{sample mean} - \text{null value}}{\text{standard error}}$$

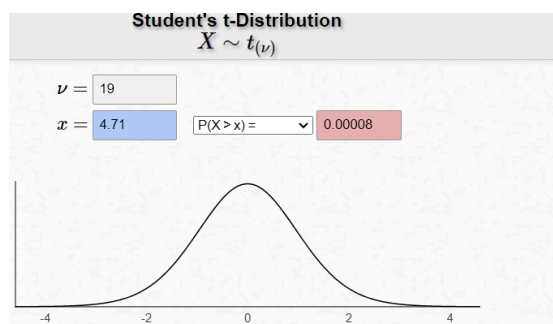
Note that this is very much like the Z-score, with one minor exception. We don't know the true population standard deviation, σ , so we estimate it with the standard deviation calculated from the 20 observed subjects in the study (this estimate is commonly denoted by s).

This t-statistic comes from what is called a t-distribution. The amount of variability in a t-distribution depends on the sample size n . Therefore, this distribution is indexed by its *degrees of freedom* (df).

Degrees of Freedom for a Single Mean (t-distribution)

For inference on a single mean, $df = n - 1$.

To find the p-value associated with this test statistic, we must remember that this is an upper-tailed test (we are trying to find evidence that the mean is greater than 45 seconds). So, the p-value will be the probability we would observe a sample mean (or a t-statistic) greater than that obtained in the actual study by chance alone, assuming the null hypothesis is true:



<https://homepage.divms.uiowa.edu/~mbognar/applets/t.html>

Note that in practice, we can use R to test our research question:

```
t_test(x = nicotine,  
       response = time_passed,  
       mu = 45,  
       alternative = "greater",  
       )
```

- ① Use the `t_test()` function from the `infer` package and designate your `x` = data set.
- ② Tell the function the `response` = variable of interest,
- ③ and the `mu` = null value (μ_0).

④ Specify the direction of the `alternative =` hypothesis.

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl>   <dbl> <chr>         <dbl>   <dbl>   <dbl>
1     4.71   19 0.0000765 greater         52.7    49.8    Inf
```

Conclusion

3. Write a conclusion in context of the problem.

At an $\alpha = 0.05$, we have evidence to conclude the population mean perceived time passed for all smokers suffering nicotine withdrawal is greater than 45 seconds ($T = 4.71$, $df = 19$, $p\text{-value} = 0.000087$).

Checking the Normality Assumption:

For the t-test to be valid, at least one of the following conditions must be met:

- Either the sample size is sufficiently large (greater than 30 or so), OR
- The distribution of the observed data is approximately normal (which would indicate that the population is normally distributed so that the Central Limit Theorem would apply even with a small sample size)

For Example 6.1, we have a sample size of 20 subjects, which is not sufficiently large. So we must check whether the data seem to come from a normal distribution.

4. Look back at the histogram of the original data. Does this seem to indicate that this is a reasonable assumption?

The histogram of the observed data is approximately normalish (one peak, somewhat symmetric). Thus, okay to use the t-distribution!

Example 6.2: Time Perception for Smokers NOT Suffering from Nicotine Withdrawal

For the data given in Example 6.1, we found evidence that the mean perceived elapsed was in fact greater than the actual 45 seconds that had elapsed. This study alone, however, doesn't really say that the nicotine withdrawal was what impaired one's perception of time. Why not?

Suppose that the researchers also studied 22 subjects who were smokers that did NOT abstain from smoking prior to the data collection (so, they were not suffering from nicotine withdrawal).

Research Question: Is there evidence the mean perceived elapsed time for all smokers **NOT** suffering from nicotine withdrawal is significantly greater than the actual 45 seconds?

```
head(nowithdrawal_data)
```

```
# A tibble: 6 × 4
  status      sex    age time_passed
  <chr>      <chr> <dbl>   <dbl>
1 no withdrawal female  38.8    52.6
2 no withdrawal female  27.3    41.4
3 no withdrawal male    45.0    51.7
4 no withdrawal female  30.0    48.0
5 no withdrawal male    32.3    52.4
6 no withdrawal female  37.5    38.0
```

Carry out the formal t-test to address this research question.

1. Set up the null and alternative hypotheses

H_0 : The population mean perceived time passed for all daily smokers NOT suffering nicotine withdrawal is equal to 45 seconds. (no impairment)

H_A : The population mean perceived time passed for all daily smokers NOT suffering nicotine withdrawal is greater than 45 seconds. (impairment)

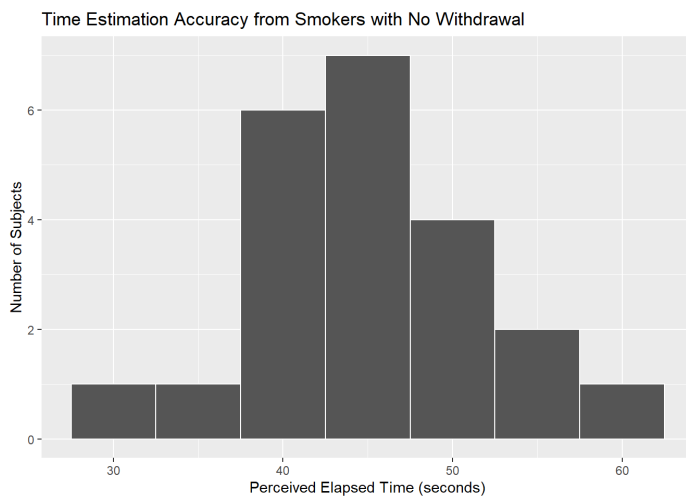
In Symbols:

- $H_0 : \mu = 45$
- $H_A : \mu > 45$

2. Check normality assumptions for using the CLT.

```
favstats( ~ time_passed, data = nowithdrawal_data)
```

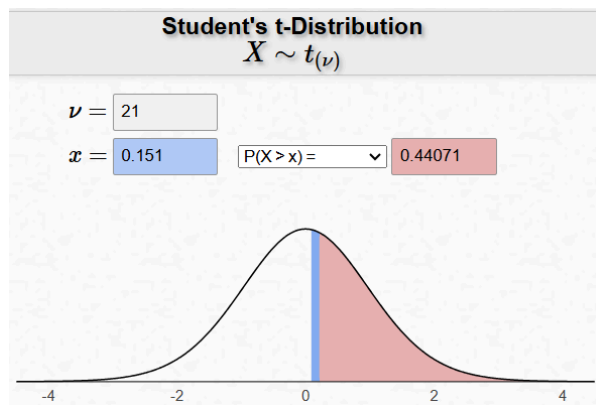
min	Q1	median	Q3	max	mean	sd	n	missing
30.86482	41.2379	45.86635	49.45927	61.21396	45.22004	6.826975	22	0



$n = 22 < 30$ – thus need to check for normality in the histogram.

Maybe slightly skewed, but decently normal. One peak, somewhat symmetric.

3. Find the t-statistic (*practice calculating this out*) and the p-value.



<https://homepage.divms.uiowa.edu/~mbognar/applets/t.html>

```
t_test(x = nowithdrawal_data,  
       response = time_passed,  
       mu = 45,
```

```
alternative = "greater"
)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl> <dbl>
1 0.151 21 0.441 greater 45.2 42.7 Inf
```

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{45.22 - 45}{6.83/\sqrt{22}} = 0.151$$

The mean of this data is only 0.151 sd above the expected mean of 45 seconds. Probably not an outlier → Fail to Reject

4. Write a conclusion in the context of the problem.

At an $\alpha = 0.05$, we do not have enough evidence to conclude the population mean perceived time passed for daily smokers NOT suffering nicotine withdrawal is greater than 45 seconds ($T = 0.151$, $df = 21$, $p\text{-value} = 0.44$).

Confidence Interval for a Single Population Mean

In **Example 6.1**, we found evidence that the mean perceived elapsed time for smokers suffering from nicotine withdrawal differed from the actual 45 seconds of time that had elapsed. Our next question is obvious: HOW MUCH does it differ? To answer this question, we must construct a confidence interval.

Recall our discussion of confidence intervals from earlier in the quarter:

This procedure does NOT require any hypotheses concerning our population parameter of interest (the population mean, in this case). We will use both our sample data (in particular, the observed mean) and what we know about the *distribution of sample means* to obtain a range of likely values for our population mean.

Warning

- A confidence interval allows us to *estimate the population parameter* of interest (recall that the hypothesis test does NOT allow us to do this). Therefore, when available, a confidence interval should always accompany the hypothesis test.
- The confidence interval does not require any hypothesized value for the population parameter. Instead, we center the confidence interval on the sample mean. Consider the following example.

Example 6.3: Estimated Perceived Time from Nicotine Withdrawal

Our goal is to construct a 95% confidence interval for the mean perceived elapsed time for smokers suffering from nicotine withdrawal (Example 6.1). To do this, we will center our distribution of sample means on the observed sample mean. Then, we will find the lower and upper endpoints that separate the middle 95% of the distribution from the rest (since we are constructing a 95% confidence interval).

Recall the general form of a confidence interval:

$$\text{point estimate} \pm \text{multiplier} \times \text{SE.}$$

Thus, the formula for calculating the endpoints of a confidence interval for a mean is given as follows:

$$\bar{x} \pm t\text{-quantile} \times \left(\frac{s}{\sqrt{n}} \right)$$

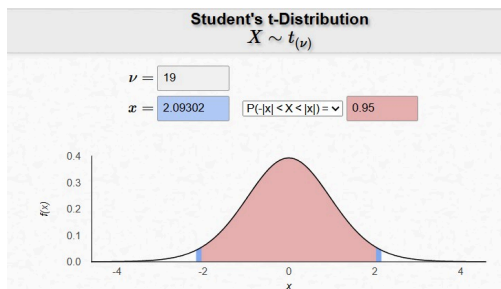
The appropriate t-quantile can be found using R (*typically I will give you a table to choose from*)
To find this value, you need the following information:

```
qt(0.975, df = 19)
```

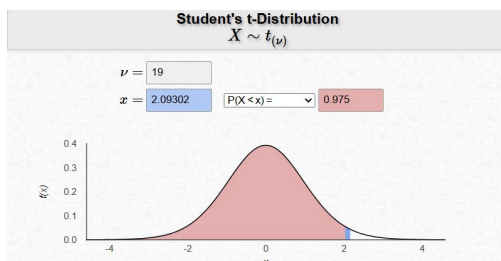
```
[1] 2.093024
```

this is our t-quantile (will be close to 1.96 for 95%)

Notice 95% falls between -2.09 and +2.09.



Thus, 97.5% falls BELOW +2.09. (2.5% in both tail ends).



- confidence level = 95%
- df = 20-1=19

Also, recall our summary statistics for time passed from Example 6.1:

```
favstats( ~ time_passed, data = nicotine)
```

```
   min      Q1   median      Q3     max    mean      sd n missing
39.11611 48.74039 53.25595 56.65965 69.36787 52.65206 7.268201 20      0
```

1. Use this information to find the endpoints of the confidence interval:

- Lower endpoint = $\bar{x} - t\text{-quantile} \times \left(\frac{s}{\sqrt{n}} \right)$

$$2.65 - 2.09 \left(\frac{7.2}{\sqrt{20}} \right) = 49.29$$

- Upper endpoint = $\bar{x} + t\text{-quantile} \times \left(\frac{s}{\sqrt{n}} \right)$

$$2.65 + 2.09 \left(\frac{7.2}{\sqrt{20}} \right) = 56.01$$

Thus, the 95% confidence interval is (49.29, 56.01).

Note that we can ask R to provide the endpoints of the 95% confidence interval for this mean when we use the `t_test` function with a `two-sided` test. We can change the confidence level with the `conf_level` argument.

```
t_test(x = nicotine,
      response = time_passed,
      mu = 45,
      alternative = "two.sided",
      conf_level = 0.95
    )
```

- ④ Change the `alternative =` to a two-sided test (CI are always two-sided),
 ⑤ and specify the `conf_level =` .

A tibble: 1 × 7

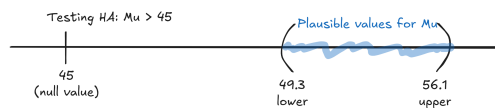
	statistic	t_df	p_value	alternative	estimate	lower_ci	upper_ci
	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	4.71	19	0.000153	two.sided	52.7	49.3	56.1

2. Interpret the meaning of this interval. What does this interval tell us about the true mean perceived elapsed time for all smokers that are suffering from nicotine withdrawal?

We are 95% confident the population mean perceived time passed for all daily smokers suffering nicotine withdrawal is between 49.3 and 56.1 seconds.

3. Does this interval agree with what you learned from the hypothesis test? Explain.

Yes, we can estimate the plausible values for the population mean are completely above 45 seconds (null value) and we rejected the null hypotheses (concluded evidence the population mean perceived time passed is greater than 45 seconds).



Key Statistical Concepts

While the main purpose of *confidence intervals* is to provide an *estimated range* of plausible values for μ , the interval should agree with conclusions found from hypothesis testing.

- If μ_0 falls within the confidence interval, then μ_0 is a plausible value for μ . Therefore, we would *Fail to Reject* the null hypothesis.
- If μ_0 falls outside the confidence interval, then μ_0 is not a plausible value for μ . Therefore, we would *Reject* the null hypothesis.

4. How would your calculations change if you wanted to obtain a 90% confidence interval, instead?

```
qt(0.90, df = 19)
```

```
[1] 1.327728
```

```
qt(0.95, df = 19)
```

```
[1] 1.729133
```

```
qt(0.975, df = 19)
```

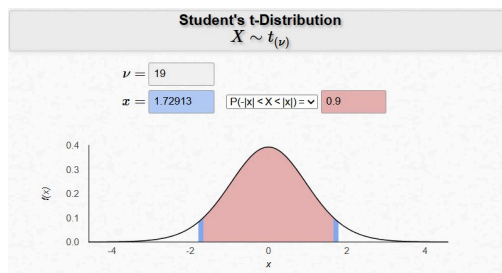
```
[1] 2.093024
```

```
qt(0.995, df = 19)
```

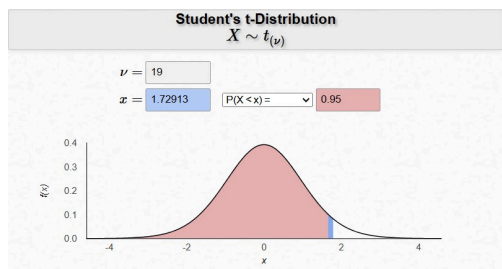
[1] 2.860935

The multiplier is now 1.73 instead of 2.09.

Notice 90% falls between -1.73 and +1.73.



Thus, 95% falls BELOW +1.73. (5% in both tail end).



References

Klein, Laura Cousino, Elizabeth J Corwin, and Michele M Stine. 2003. "Smoking Abstinence Impairs Time Estimation Accuracy in Cigarette Smokers." *Psychopharmacology Bulletin* 37 (1): 90–95.